

A Pocket Offline Model for Simultaneous Speech Translation as CUNI

Submission to IWSLT 2026

Aziz Sharipov Ortega
Charles University, MFF
azdisharipov@gmail.com

Dominik Macháček
Charles University, MFF, ÚFAL
& University of Edinburgh
machacek@ufal.mff.cuni.cz

Abstract

We implement a direct speech translation model Canary for simultaneous translation with AlignAtt simultaneous policy. We focus on Nemo toolkit with the recent state-of-the-art foundation model Canary-1B-v2 that has only one billion of parameters, which is suitable for small pocket devices. This is a CUNI submission to IWSLT 2026 Simultaneous Speech Translation Shared task on Czech to English and English to German and Italian.

The strengths of our system are: (1) high translation quality, overperforming similarly sized baselines both in low- and high-latency regimes in both computationally unaware and aware simulations; (2) low computational requirements, as the neural model that we use has only 1B parameters; (3) multilinguality – support of 25 source and 25 target languages.

1 Introduction

In this paper, we describe a submission of the Charles University (CUNI) to IWSLT 2026 Simultaneous Speech Translation Task (Anastasopoulos et al., 2026). Our system is built on top of Canary-1B-v2¹, even though technically *Canary* is a family of models (Sekoyan et al., 2025) with AlignAtt (Papi et al., 2023) simultaneous policy. Following SimulStreaming (Macháček and Polák, 2025), the highest performing submission in IWSLT 2025 (Abdulmumin et al., 2025), we also use Silero Voice Activity Detection (VAD) in the pipeline to reduce noise and hallucination in non-speech periods of audio. We validate the latency of our systems in computationally unaware simulation, and submit also a version for computational aware simulation.

One of the strengths of our system is high multilinguality. The Canary model supports 25

source and 25 target languages for direct translation, which overlaps with three language pairs of IWSLT 2026 we focus on: Czech to English, English to German, and English to Italian. The model is also very small and high quality. It has only 1B parameters, making it operational even on small mobile devices.²

We follow the approach of repurposing offline speech translation models for simultaneous mode because it has been shown as a simple, effective, and well-performing strategy, e.g. in Papi et al. (2022) and in Macháček and Polák (2025). The benefits of high quality, multilinguality, flexibility, and robustness often overweight the limitations such as hallucination on partial source sentences, not using left-only context, or repeated encoding of the whole source buffer with every new incoming chunk, that can be avoided by specific simultaneous methods, which usually require costly training, e.g. for stability on source prefixes (Niehues et al., 2018), for learning to wait or translate (Koshkin et al., 2026), or specific simultaneous models (Ariavazhagan et al., 2019; Labiausse et al., 2025). We observe a higher demand for offline speech translation than for simultaneous. New, higher quality and diverse offline models are becoming available, but there is a gap in flexible and robust implementations that enable processing any offline model in simultaneous mode. This is a gap that we aim to fill with this work.

Our primary goal is therefore to pioneer Canary usage in simultaneous mode and evaluate it against state-of-the-art systems, to test whether Canary in simultaneous mode can be used in practical applications and as a baseline in future research.

We conclude we have reached the main goal. The results show competitive and in some cases

¹Throughout this paper, we use a simple term *Canary* to refer to the *Canary-1B-v2* model

²<https://www.siliconflow.com/articles/en/best-lightweight-LLMs-for-mobile-devices> suggests that even 8B-parameter models are suitable for mobile devices.

even superior performance to state-of-the-art baselines while being much smaller in size. Our results show improvements by over 4-5 BLEU points over the organizers’ baseline on English to German and Italian and 5-8 BLEU points for Czech to English over the IWSLT 2025 best performing system (Macháček and Polák, 2025).

The paper is structured as follows: in the Section 2, we introduce the core model and policies and provide a relevant background. In Section 3, we dive into implementation details of our submission, while Section 4 serves as a detailed description of the frameworks and evaluation metrics used, which we then follow by the evaluation result in Section 5. Finally, we wrap up the paper with a conclusion and limitations regarding the implementation.

2 Background

Canary-1B-v2 (Sekoyan et al., 2025) is a strong multitask speech transcription and translation model with state-of-the-art performance. It outperforms *whisper-large-v3* (Radford et al., 2022) and *seamless-m4t-medium* in automatic speech recognition (ASR) and automatic speech translation (AST) tasks, and on some datasets is even superior to *seamless-m4t-v2-large*, which is twice as large in size. Additionally, it supports injecting context in the decoder prompt to bias the prediction with in-domain context, and timestamping. Unlike Whisper, Canary does not need to work with fixed size audio inputs and was instead trained to support audios in 0 to 40 seconds range, which makes it a perfect candidate for simultaneous adaptation.

AlignAtt (Papi et al., 2023) is a simultaneous policy. This policy idea is to use the cross-attention layer in the decoder to omit the suffix of hypothesis after the first token that attends to a predefined frame threshold. It has been shown to work well in long-form translation (Papi et al., 2024) and with Whisper (Wang et al., 2024; Macháček and Polák, 2025), but has not been previously applied on Canary.

Sliding window (Sen et al., 2022) is another simultaneous policy. Its core idea is to re-translate a window of audio input, use longest common subsequence with the previous translation to avoid repeating, and sliding the window with newly available audio chunk. Although being outperformed by AlignAtt, the only available implementation of Ca-

nary in simultaneous mode by Gaido et al. (2025) uses this approach. We use it as a baseline for comparison to our system.

3 Implementation

Let us describe the details of the implementation of our systems and the end-to-end long-form speech-to-text processing pipeline.

Simulation frameworks There are two frameworks for end-to-end simulation of speech-to-text simultaneous systems on pre-recorded audio that also allow integration of new models: the earlier introduced is SimulStreaming (Macháček and Polák, 2025) and the latter Simulstream (Gaido et al., 2025). We use primarily SimulStreaming because it provided a robust implementation of SileroVAD (Team, 2024), a streaming voice activity detection that filters non-voice parts of the input audio and allows to save computational power on processing empty input, while also avoiding potential hallucinations.

Since IWSLT 2026 task required Simulstream for computational aware evaluation, we finally transfer our Canary implementation to Simulstream to enable it. However, in case of differences, we consider our SimulStreaming implementation as the primary.

Adapting Nemo It was not straightforward to directly apply the AlignAtt policy to Canary model in Nemo, since the API did not support decoder forced prefix³ injection required for the incremental output. Hence, we highlight our contribution to the Nemo’s speech framework (Harper et al.) that allows for this core incremental concept to be used with Canary. With our contribution, now the prefix can be provided as an optional initial prompt to the decoder. Additionally, we fixed the bug in the cross-attention outputs for the beam decoder strategy, which allowed us to get deterministic dimension outputs and map the cross-attention scores to their respective output tokens.

Processing loop Our audio processing pipeline follows the stages of prototypical systems described in Papi et al. (2025a), Section 3. We keep the same numbering of steps:

1.-2. Audio acquisition and segmentation using

³The approach is described clearly in this thread <https://github.com/openai/whisper/discussions/117#discussioncomment-3727051>

VAD is identical to SimulStreaming. Refer to Macháček and Polák (2025), Section 3.

3. Speech buffer update. The incoming chunk, which has *MinChunkSize*⁴ seconds if the end of voice is not detected, or less otherwise, is concatenated with the speech buffer.
4. Hypothesis generation. Canary is an attention based encoder-decoder (AED) model. When audio is inputted, it is first encoded by the encoder. Then an initial prompt is passed to the decoder, which contains information about the source and target languages, decoder forced-prefix if any is in the buffer, as well as some other special tokens that affect final output, but are not relevant in the context of implementation, for example, whether to use timestamps. Then the model decodes the target as long as the AlignAtt policy allows. If any part of a word is inside the AlignAtt’s frame threshold, the word is removed from the output. If the current chunk is not final, the decoding continues until the most attended source frame is close to the end of the audio, which is indicated by the *Frames* parameter. In case the current chunk is final, we do not apply AlignAtt and output the whole generated sequence.
5. Audio and context buffers. We use the following two buffers: (1) source audio buffer, which accumulates latest 30 seconds of audio in raw form. We note that although storing the mel-features instead of raw form is more efficient, as it skips the additional preprocessing step on every new chunk, we implement a simpler buffer as we were more focused on the computationally unaware scenario, and (2) forced decoding target buffer that contains the stable part of the hypothesis that was decoded from current audio buffer.

If the audio buffer has length of 30 seconds or more, we remove the first speech chunk from the source audio buffer. At the same time, we discard the text that was decoded with the first chunk from forced decoding. After shifting the buffer, it may happen that the audio is not entirely parallel to the forced decoded target buffer, but the results show that the model

performs well with that. We leave possible improvement for further work.

4 Development

Dev sets For English-German and English-Italian directions, we use the MCIF (Papi et al., 2025b) dataset, as provided by the IWSLT 2026 organizers. For Czech-to-English direction, we use the IWSLT 2026 dev set, which consists of meetings of the Czech Chamber of Deputies.

MT metrics In our initial development, we relied on BLEU (Papineni et al., 2002) and ChrF (Popović, 2015) as our guiding quality scores with more emphasis on ChrF. However, for selecting the final system candidates, we used COMET-XL (Guerreiro et al., 2024) as it is the main MT quality metrics for IWSLT 2026.

Latency metrics We use computationally unaware LongYAAL (Polák et al., 2025) to select best hyper-parameters in the low- and high-latency ranges (2 and 4 seconds respectively).

5 Results

We evaluate our systems on the MCIF dev set for English-to-German and English-to-Italian, and on the IWSLT 2026 dev set for Czech-to-English. We compare against the following baselines: (1) the organizers’ cascade baseline, which consists of a local agreement ASR component followed by a neural MT model (*Qwen3-ASR-1.7B* (Shi et al., 2026) and *Qwen3-4B-Instruct-2507* (Yang et al., 2025) respectively), (2) the Canary sliding window system as implemented in Simulstream, (3) Canary in offline mode, and finally (4) Simul-Streaming for Czech to English direction.

Metric scores and comparison Table 2 reports our main results for the English-to-German and English-to-Italian directions. Our Canary with AlignAtt system outperforms the organizers’ baseline across all four configurations – high and low latency for both language pairs – on BLEU, chrF, and XCOMET-XL. The improvements are most pronounced in the high-latency regime, where we gain over 4 BLEU points on English-to-German and more than 6 BLEU points on English-to-Italian. Notably, this holds even when comparing against the strongest organizers’ baseline configuration that uses transcript context – our system outperforms it on all quality metrics in the high-latency regime for both language pairs. In the low-latency regime, the

⁴We mark system parameters with italics.

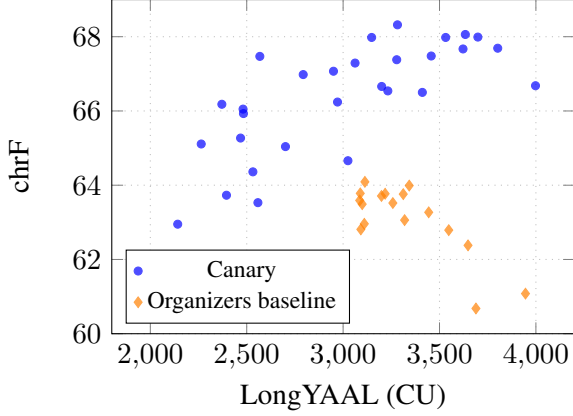


Figure 1: English-to-Italian dev chrF vs. LongYAAL (CU) for Canary vs. the organizers baseline. Each blue point represents one Canary candidate from the search of *MinChunkSize* and *Frame* parameters. Orange points represent the organizers baseline system with grid-search for *segment-length* and *step-length*

quality gains are more modest on BLEU and chrF, especially for English to German direction, where the system falls behind the baseline, but XCOMET-XL consistently favors our system.

Figure 1 further illustrates the quality-latency trade-off on English-to-Italian. Notice that since our system has two parameters that influence latency-quality without any clear relation, it is not possible to plot the system candidate as a line, as it is typically done with other policies. Our Canary system dominates the organizers’ cascade baseline across the entire latency range, achieving higher chrF at comparable or lower latency.

Table 2 additionally reports the scores of the Simulstream authors’ Canary sliding window implementation, using the configuration they provide as default (chunk 2s, window length 12, matching threshold 0.1). Our AlignAtt-based system substantially outperforms this re-translation approach: on English-to-German, we gain over 8 BLEU points in the high-latency regime, and on English-to-Italian, we gain over 7 BLEU points at a lower latency. This confirms that, while the sliding window policy is a practical and simple way to repurpose Canary for simultaneous mode, AlignAtt provides a clearly superior quality-latency tradeoff.

For the Czech to English direction, we compare our system against the SimulStreaming baseline, which was the top-performing system in IWSLT 2025. As shown in Table 2, Canary with AlignAtt substantially outperforms this baseline in both latency regimes. In the high-latency setting, our

Chunk	Frame	BLEU	chrF	Latency
<i>Latency < 2s</i>				
0.5	12	27.78	56.68	1997
2.5	1	23.86	53.72	1991
1.0	5	22.67	53.35	1652
1.0	4	20.13	51.85	1390
1.5	4	20.07	51.83	1899
<i>Latency < 4s</i>				
2.5	20	32.01	59.14	3641
3.5	16	31.90	59.26	3921
2.0	20	31.69	59.24	3240
3.5	12	31.43	59.01	3968
3.0	12	31.36	58.95	3954

Table 1: Top-performing Canary cs-en candidates for different latency regimes. We report BLEU (the higher, the better), chrF, and LongYAAL (CU) latency on the cs-en dev set.

system improves BLEU by nearly 8 points and achieves large gains in chrF as well. In the low-latency regime, we observe a similarly strong improvement of more than 5 BLEU points.

Finally, in Table 2 we report the results of running the model on the dev set in the offline mode. Canary’s offline inference on long audios works as described in (Sekoyan et al., 2025) in Section 6.4.1. We show that our approach works better than the offline mode and can be used instead if the running time is not a concern.

Grid-search We perform grid search to find the optimal *MinChunkSize*, *Frames* to meet the high-latency threshold of the IWSLT 2026 Simultaneous task, which is below LongYAAL 2000ms, and the high-latency threshold below 4000ms LongYAAL, both in computationally unaware modes. For that, we used the Omisteval(Polák et al., 2025) tool, the same tool used by the shared task organizers to evaluate the submissions.

6 Conclusion

We presented a compact and practical approach to simultaneous speech translation based on the offline Canary-1B-v2 model and the AlignAtt policy. Our main contribution is an end-to-end implementation that adapts a strong offline speech translation model to simultaneous use in the Nemo ecosystem, together with the necessary support for forced-prefix decoding and cross-attention-based truncation.

The results show that this repurposing approach

		Reg.	BLEU	chrF	XCOMET-XL	LongYAAL (ms)
EnDe	baseline organizers (no ctx)	H	27.44	59.66	0.8351	3431
	baseline organizers (ctx)	H	27.66	59.92	0.8428	3353
	baseline organizers (ctx)	L	22.59	57.51	0.7651	1747
	Canary ours	H	31.73	60.83	0.8776	3761
	Canary ours	L	20.70	52.60	0.7744	1677
	Canary sliding window	H	23.65	58.53	–	2925
	Canary offline	–	25.01	55.52	–	–
EnIt	baseline organizers (no ctx)	H	37.28	65.44	0.7806	3300
	baseline organizers (ctx)	H	37.76	65.77	0.7877	3231
	baseline organizers (ctx)	L	31.45	63.03	0.6960	1735
	Canary ours	H	43.56	68.32	0.8227	3282
	Canary ours	L	34.79	62.21	0.7618	1972
	Canary sliding window	H	35.52	66.07	–	2724
	Canary offline	–	36.78	62.22	–	–
CsEn	SimulStreaming Whisper	H	24.20	50.36	–	3512
	Canary ours	H	32.01	59.26	0.8133	3641
	Canary ours	L	27.78	56.68	0.7633	1997
	SimulStreaming Whisper	L	22.11	49.55	–	1804

Table 2: Dev set results of Canary with AlignAtt for simultaneous translation compared to baselines. Reg. indicates latency regime: H = high (<4s), L = low (<2s). LongYAAL is reported in milliseconds; lower is better. Our system is shaded. Baseline organizers entries differ by use of transcript context (ctx) or not (no ctx).

is effective in practice. Across the evaluated language pairs, Canary with AlignAtt achieves competitive quality and in several settings clearly outperforms the organizers’ baselines and the previously available Canary sliding-window implementation. At the same time, the system remains lightweight, with only 1B parameters, and supports a relatively broad multilingual setup, which makes it attractive for deployment in resource-constrained scenarios.

Limitations

We also report that during our experiments with the decoder prompt and context, we have not found an optimal setup to bias in-domain predictions. Injecting both forced prefix and context makes the model stall and not produce any output. Most likely, this has to do with the fact that Canary-1B-v2 was not trained for such a set of inputs, which makes it confused and nothing is outputted inference time.

References

- Idris Abdulmumin, Victor Agostinelli, Tanel Alumäe, Antonios Anastasopoulos, Luisa Bentivogli, Ondřej Bojar, Claudia Borg, Fethi Bougares, Roldano Cattoni, Mauro Cettolo, Lizhong Chen, William Chen, Raj Dabre, Yannick Estève, Marcello Federico, Mark Fishel, Marco Gaido, Dávid Javorský, Marek Kasztelnik, and 33 others. 2025. [Findings of the IWSLT 2025 evaluation campaign](#). In *Proceedings of the 22nd International Conference on Spoken Language Translation (IWSLT 2025)*, pages 412–481, Vienna, Austria (in-person and online). Association for Computational Linguistics.
- Antonios Anastasopoulos and 1 others. 2026. Speech translation and metrics in 2026: Findings of the IWSLT campaign. In *Proceedings of the 23rd International Conference on Spoken Language Translation (IWSLT 2026)*. Association for Computational Linguistics.
- Naveen Arivazhagan, Colin Cherry, Wolfgang Macherey, Chung-Cheng Chiu, Semih Yavuz, Ruoming Pang, Wei Li, and Colin Raffel. 2019. [Monotonic infinite lookback attention for simultaneous machine translation](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 1313–1323, Florence, Italy. Association for Computational Linguistics.
- Marco Gaido, Sara Papi, Mauro Cettolo, Matteo Negri, and Luisa Bentivogli. 2025. [Simulstream: Open-source toolkit for evaluation and demonstration of streaming speech-to-text translation systems](#). Preprint, arXiv:2512.17648.
- Nuno M. Guerreiro, Ricardo Rei, Daan van Stigt, Luisa Coheur, Pierre Colombo, and André F. T. Martins. 2024. [xCOMET: Transparent machine translation evaluation through fine-grained error detection](#). *Transactions of the Association for Computational Linguistics*, 12:979–995.
- Eric Harper, Somshubra Majumdar, Oleksii Kuchaiev, Li Jason, Yang Zhang, Evelina Bakhturina, Vahid Noroozi, Sandeep Subramanian, Koluguri Nithin, Huang Jocelyn, Fei Jia, Jagadeesh Balam, Xuesong Yang, Micha Livne, Yi Dong, Sean Naren, and Boris

- Ginsburg. [NeMo: a toolkit for Conversational AI and Large Language Models](#).
- Roman Koshkin, Je Haesung, Lianbo Liu, Hao Shi, Meng Zhao, Yusuke Fujita, and Yui Sudo. 2026. [Streaming translation and transcription through speech-to-text causal alignment](#).
- Tom Labiausse, Laurent Mazaré, Edouard Grave, Patrick Pérez, Alexandre Défossez, and Neil Zeghidour. 2025. [High-fidelity simultaneous speech-to-speech translation](#). *Preprint*, arXiv:2502.03382.
- Dominik Macháček and Peter Polák. 2025. [Simultaneous translation with offline speech and LLM models in CUNI submission to IWSLT 2025](#). In *Proceedings of the 22nd International Conference on Spoken Language Translation (IWSLT 2025)*, pages 389–398, Vienna, Austria (in-person and online). Association for Computational Linguistics.
- Jan Niehues and 1 others. 2018. [Low-Latency Neural Speech Translation](#). In *Proc. Interspeech 2018*.
- Sara Papi, Marco Gaido, Matteo Negri, and Luisa Bentivogli. 2024. [StreamAtt: Direct Streaming Speech-to-Text Translation with Attention-based Audio History Selection](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand.
- Sara Papi, Marco Gaido, Matteo Negri, and Marco Turchi. 2022. [Does simultaneous speech translation need simultaneous models?](#) In *Findings of the Association for Computational Linguistics: EMNLP 2022*, pages 141–153, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Sara Papi, Peter Polák, Dominik Macháček, and Ondřej Bojar. 2025a. [How “real” is your real-time simultaneous speech-to-text translation system?](#) *Transactions of the Association for Computational Linguistics*, 13:281–313.
- Sara Papi, Marco Turchi, and Matteo Negri. 2023. [Alignatt: Using attention-based audio-translation alignments as a guide for simultaneous speech translation](#). *Preprint*, arXiv:2305.11408.
- Sara Papi, Maike Züfle, Marco Gaido, Beatrice Savoldi, Danni Liu, Ioannis Douros, Luisa Bentivogli, and Jan Niehues. 2025b. [Mcf: Multimodal crosslingual instruction-following benchmark from scientific talks](#). *Preprint*, arXiv:2507.19634.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. [Bleu: a method for automatic evaluation of machine translation](#). In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318, Philadelphia, Pennsylvania, USA. Association for Computational Linguistics.
- Peter Polák, Sara Papi, Luisa Bentivogli, and Ondřej Bojar. 2025. [Better late than never: Meta-evaluation of latency metrics for simultaneous speech-to-text translation](#). *Preprint*, arXiv:2509.17349.
- Maja Popović. 2015. [chrF: character n-gram F-score for automatic MT evaluation](#). In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395, Lisbon, Portugal. Association for Computational Linguistics.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. [Robust speech recognition via large-scale weak supervision](#). *arXiv preprint*.
- Monica Sekoyan, Nithin Rao Koluguri, Nune Tadevosyan, Piotr Zelasko, Travis Bartley, Nikolay Karpov, Jagadeesh Balam, and Boris Ginsburg. 2025. [Canary-1b-v2 & parakeet-tdt-0.6b-v3: Efficient and high-performance models for multilingual asr and ast](#). *Preprint*, arXiv:2509.14128.
- Sukanta Sen, Ondřej Bojar, and Barry Haddow. 2022. [Simultaneous translation for unsegmented input: A sliding window approach](#). *Preprint*, arXiv:2210.09754.
- Xian Shi, Xiong Wang, Zhifang Guo, Yongqi Wang, Pei Zhang, Xinyu Zhang, Zishan Guo, Hongkun Hao, Yu Xi, Baosong Yang, Jin Xu, Jingren Zhou, and Junyang Lin. 2026. [Qwen3-asr technical report](#). *Preprint*, arXiv:2601.21337.
- Silero Team. 2024. Silero vad: pre-trained enterprise-grade voice activity detector (vad), number detector and language classifier. <https://github.com/snakers4/silero-vad>.
- Haoyu Wang, Guoqiang Hu, Guodong Lin, Wei-Qiang Zhang, and Jian Li. 2024. [Simul-whisper: Attention-guided streaming whisper with truncation detection](#). In *Interspeech 2024*, pages 4483–4487. ISCA.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 41 others. 2025. [Qwen3 technical report](#). *Preprint*, arXiv:2505.09388.